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Question 9

Table 1 reports the optimal tuning parameters and out-of-sample accuracy for each of the five machine learning algorithms. The decision tree performed the best with an accuracy of 87%, followed by logistic regression and SVM both at 85%. The neural network and kNN performed similarly at 84%. Overall, the tree model appears to be the best classifier for predicting whether someone is a high earner.

Table 1: Machine Learning Algorithm Comparison

Algorithm	**Best Tuning Parameters**	**Out-of-Sample Accuracy**
Logistic Regression	$\lambda = 0.00$	0.85
Decision Tree	cost_complexity = 0.00, tree_depth = 20, min_n = 15	0.87
Neural Network	hidden_units = 1, penalty = 4.00	0.84
kNN	neighbors = 30	0.84
SVM	cost = 1.00, rbf_sigma = 0.25	0.85